Rev0 PCB Axial-Flux Motor: Maxwell Modelling, Optimisation and Final Build Report

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Executive summary

The Rev0 motor is locked at a **1.0 mm air gap and a 2.0 mm steel back plate**. Its final model, REV0_FINAL_G100_T20_FROZEN.aedt, is built and validated in AEDT 2025 R2. Provisional performance is **18.0 mN·m/A per phase**, about **27 mN·m per A peak** of three-phase torque, and **1.34 V rms per phase at 1000 rpm**. One validation check is still open: the three-phase synchronous check.

- **Detailed PCB model.** The single board is electrically validated: R = 0.2777 Ω per board-phase across all three phases (0.19% spread), with L ≈ 2.07 µH.
- **Rotor reference model.** The one-board rotor model gives K1 = 6.09 mN·m/A per board-phase at a 1 mm gap. The waveform repeats over the full period to 0.02%.
- **Five-board stator.** It was built board by board and frozen. A 2-board solve then proved current flows through the solder-bridge junctions: R = 0.553 Ω against 0.555 Ω expected.
- **Optimisation used 3 extra solves instead of a 12-hour sweep.** Doubling the plate from 2 mm to 5 mm adds only 0.1% torque. The field from the magnets falls off at 0.303 /mm (about 26% per mm of distance from the magnets). That decay rate lets the five boards be added together board by board.
- **The stack gives 2.96× one board, not 5×.** The board furthest from the rotor sees only 30% of the field that board 1 sees.
- **Rev1 (S1–R1–S2–R2) is specified and scripted.** It needs one more single-board solve before its numbers can be computed.

All figures are provisional until the three-phase check (REF_ROTOR_MS_3PH) passes and the 0.5 mm point confirms the decay law.

Scope and ground rules

This report covers modelling, verifying and optimising the Rev0 PCB axial-flux motor in ANSYS Maxwell 3D (AEDT 2025 R2), and specifying Rev1.

- **Source of truth:** PCB_MAXWELL_SOURCE_PACKAGE and the Rev0 specification. Older JSON files and scripts are aids only. Any conflict is raised, never silently resolved.
- **Division of work:** Olwethu defines the motor and makes every engineering selection. Claude builds the models, automates AEDT and processes the results.
- **Hardware limit:** the lab PCs have 16 GB of RAM. Two runs froze them at Maxwell's inductance-matrix step, and one crashed during a long sweep. So the full motor is never solved in one piece.
- **Evidence tags used below:**
  - **[AEDT]**: solved or checked in Maxwell.
  - **[Py]**: a Python calculation on AEDT data, or a geometry check.
  - **[Est]**: an analytical estimate.
  - **[Assump]**: an assumption.

The Rev0 motor

Rev0 is a single-sided axial-flux machine: a stack of five PCB stator boards facing one rotor with 16 magnets on a steel back plate. There is no iron in the stator.

Part	Specification
Order along the shaft	5-PCB stator → 1.0 mm air gap → 16 magnets → steel back plate
PCB	12-sided, ~93 mm across flats, 25 mm centre opening; 6 × 1 oz (35 µm) copper layers; 12 × 3.1 mm holes on a 90 mm PCD
Winding	Gould mini_wave-bb, three phases; per phase 148 copper islands and 205 vias form one path, In on L1 and Out on L6
Board build	1.0 mm board + 0.01016 mm solder mask each face = 1.02032 mm pitch; boards touch mask to mask

Part	Specification
Stack wiring	Per phase, board k L6 Out is joined to board k+1 L1 In by a rim solder bridge, giving 5 boards in series; leads only on the string ends; star point not yet made
Terminal positions (In/Out pads)	A (−22.75, 39.40); B (0, 45.5); C (22.75, 39.40) mm
Magnets	16 × N35, 20 × 6 × 2 mm, axially magnetised, alternating N/S, centre radius 28 mm
Back plate	steel_1010, 86 mm OD, 19 mm bore, **2.0 mm** (selected)
Shaft	19 mm steel_1010 (assumed material)

The phases are rotated copies of each other. Phase B is Phase A turned −30° mechanical and Phase C is A turned −60° (≥ 99.96% copper overlap) [Py]. With 8 pole pairs that is A 0°, B −120°, C +120° electrical.

Modelling method

The motor's performance comes from magnetostatic solves of one detailed board at several rotor angles and distances. The five boards are then added together board by board. A full five-board rotating solve does not fit in 16 GB.

1. **Magnetostatic, not transient.**
   - A detailed T-Omega transient takes over 25 min per time step for one phase, and Transient A-Phi cannot model motion in AEDT 2025 R2.
   - Magnetostatic solves at fixed rotor angles take 11–20 min each.
   - The inductance matrix is switched off, because that step froze the lab PCs twice.
2. **FR-4 and solder mask modelled as air.** Neither is magnetic or conductive, so they don't affect a magnetostatic solve. A check run moved R and the stored energy by at most 0.22% [AEDT]. Board geometry, pitch and air gap are unchanged.
3. **Torque reciprocity gives the back-EMF without a transient.** The rate of change of flux linkage with rotor angle, $d\lambda/d\theta$, equals T/I. Back-EMF is $e = \omega \cdot d\lambda/d\theta$.
   - The torque is taken as the Lorentz force on the copper, which is linear in current. The rotor's virtual torque carries a numerical bias of about ±0.5 mN·m, so it is only a cross-check.

- In the ±1 A smoke runs, the copper and rotor torques agreed within 0.05%.

4. **Ideal three-phase torque** = 1.5 · K1 · I_peak, assuming the current is in phase with the back-EMF.
5. **Board-by-board stack sum.** The copper and FR-4 are non-magnetic, so the rotor's field doesn't depend on where the boards sit. Board k sees the rotor at d_k = g + (k−1) · 1.02032 mm. The stack constant is the phasor sum of the single-board results.

```
K_{stack}(g) = \left| \sum_{k=1}^{5} K_1(d_k)\, e^{j\psi(d_k)} \right|,
\qquad K_1(d) = K_1(1\,\mathrm{mm})\, e^{-k_{eff}(d-1)}
```

6. **Decay law.** The exponential follows from the analytical model for coreless axial-flux machines (Vatani et al., UK SPARK Lab, reported within 1–2% of 3D FEA). Its single-sided limit gives a field that falls as exp(−πz/τ). At r = 28 mm the pole pitch τ = 11.0 mm, so π/τ = 0.286 /mm. The decay rate k_eff is fitted to the FEM points rather than assumed.
7. **Figure of merit.**

```
K_m = \frac{1.5\,K}{\sqrt{1.5\,R_{\phi}}}
```

Km is torque per square root of copper loss, independent of current. For Rev0, R_φ = 5 × 0.2777 = 1.389 Ω, excluding the external leads.

Scripted workflow

Every model is built and solved by the REV0_PIPELINE scripts (v9.1), designed so that a frozen PC loses at most one solve.

1. **Stages.** project → stator → rotor → region/band → excitation → solver setup → validate → analyze → export. Each stage that succeeds is saved as a checkpoint, and a run can resume from any stage.
2. **Stepwise sweep.** Each rotor angle is solved, saved and logged before the next one starts, and the project is reopened between angles to free memory. A crashed sweep resumes at the next unsolved angle. This came from the 2026-10-05 baseline crash, where Optimetrics saved nothing until the end.
3. **Build first, then test.**
   - The five-board stator was built from an empty project, one board at a time. Each board was validated, saved and checkpointed before the next.
   - The boards were then connected, the junctions probed, and the result frozen.

- Tests and final builds only ever open *copies* of a frozen file.

4. **Restartable batches across PCs.**
   - Scripts and results live in the shared OneDrive folder `Motor run`, while each PC solves on its own local disk. Solving inside OneDrive risks sync locks on multi-GB solver files.
   - When a run finishes, its summaries are copied to `Motor run\results`. A run already finished on any PC is skipped everywhere.
5. **Acceptance checks in every run.**
   - AEDT validation.
   - The expected air gap and phase rotations.
   - Junction body probes.
   - The region's +Y face must sit on the lead ends, so the coil terminals are legal.
   - The solve is rejected on any error-level message or missing solution.

Verification of the detailed model

Every check on the single board and the rotor reference model has passed. The resistance, inductance and back-EMF figures agree across independent methods to within 1%.

Check (date)	Model	Result	Status
Gate 1, 1 A DC per phase (10-04)	1 board, magnetostatic	R_A / R_B / R_C = 0.27776 / 0.27736 / 0.27789 Ω; spread 0.19%; +2.9% against the 0.2699 Ω reference	Passed [AEDT]
Gate 2A, T-Omega (10-05)	1 board, Phase A, 2 steps	L ≈ 2.07 µH from both flux linkage and induced voltage, which agree within 1%	Passed [AEDT]
Board, all three phases (10-05)	FR-4 and mask solid, then as air	R within 0.5% of Gate 1; air substitution moves R/energy ⩽ 0.22%; L − M ≈ 1.90 µH	Passed [AEDT]

Check (date)	Model	Result	Status
Rotor smoke test, 0°/11.25° ± 1 A (10-05)	1 board + rotor	Copper and rotor torque agree within 0.05%; copper torque reverses exactly with current (to 0.002%)	Passed [AEDT]
Baseline sweep, 0–45° ± 1 A (10-06)	1 board + rotor, 1 mm gap, 2 mm plate	**K1 = 6.09 mN·m/A**, ψ = −30.2° elec, λ1 = 0.7615 mWb; repeats over the full period to 0.02%; half-wave antisymmetry 0.05%	Passed [AEDT]
Phase rotations (10-05)	Gerber geometry	B = A rotated −30°, C = A rotated −60° mech (≥ 99.96% overlap)	Passed [Py]

More from the baseline sweep:

- Back-EMF per board-phase is 0.638 V peak / 0.451 V rms at 1000 rpm.
- The third harmonic is 11.8% of the fundamental. In a star connection the third harmonic cancels between phases.
- Cogging stays within the ±0.5 mN·m numerical noise and shows no repeatable pattern.
- Steel peak |B| is 1.39–1.44 T and the mid-gap mean |Bz| is 0.168 T.

Five-board stator and junction proof

The five-board Rev0 stator is built, frozen as REV0_STATOR_FROZEN.aedt, and electrically proven. A 2-board solve through one solder bridge measured **0.553 Ω against 0.555 Ω expected** [AEDT].

- **Build.**
  - Started from an empty project and built PCB1…PCB5 one at a time, each validated and checkpointed.
  - Per phase, a 1 × 1 mm copper bridge spans the 0.02032 mm mask-to-mask gap from board k's L6 Out pad to board k+1's L1 In pad.
  - The boards are separate touching conductors; uniting them into one solid crashed AEDT in v7.
  - Raised leads are on the string ends only.

- **Geometry proof.** At every junction AEDT reported the expected conductor in the board k pad, the bridge and the board k+1 pad: 12/12 junctions passed (36 probes) [AEDT].
- **Fix during the build.** Maxwell first rejected the six terminals as "illegal external terminal", because the region's +Y padding was 5 mm. With the padding set to 0, the lead faces sit on the boundary. The pipeline now enforces this.
- **Electrical proof (STACK_JUNCTION_TEST, 2 boards, 1 A DC).**
  - The solve was accepted with no conduction-path error.
  - Ohmic loss: PCB1 0.27654 W, PCB2 0.27659 W, bridge 3.6 µW.
  - Both boards carry the full current, so it passes through the bridge with no short or bypass.
- **Inductance from the same run (estimate).** The stored energy gives L ≈ 6.23 µH for the two boards in series. With L ≈ 2.07 µH per board, that implies a board-to-board mutual inductance of about 1.0 µH (coupling ≈ 0.5). It comes from a 2-pass solve and needs confirming before use [Est].

Back plate thickness: 2 mm is enough

Going from a 2 mm to a 5 mm steel back plate raises K1 by only **0.1%** (6.09 → 6.097 mN·m/A) and adds about 130 g. So 2 mm was selected [AEDT, REV0_DOE_BI50].

Rotor angle (mech)	dλ/dθ, 2 mm plate (mWb/rad)	dλ/dθ, 5 mm plate (mWb/rad)
0°	+5.317	+5.322
5.625°	+1.018	+1.018
11.25°	−2.347	−2.349
16.875°	−6.359	−6.364

- **The extra steel adds no air-gap field.** The mid-gap mean |Bz| is 0.168 T for both plates, and ψ is unchanged at −30.2°.
- **The steel peak of 1.35–1.41 T is a local hot spot.** It barely changed with 5 mm, so it sits at the magnet corners and the plate's bulk isn't saturated. The torque comparison is the measure that matters.
- **Selection rule:** the thinnest plate within 1–2% of the "plenty of iron" torque. 2 mm passes with a wide margin.

- **Mass:** about 43.5 g per mm of plate thickness (86 mm OD, 19 mm bore). That makes 2 mm ≈ 87 g against 5 mm ≈ 218 g.

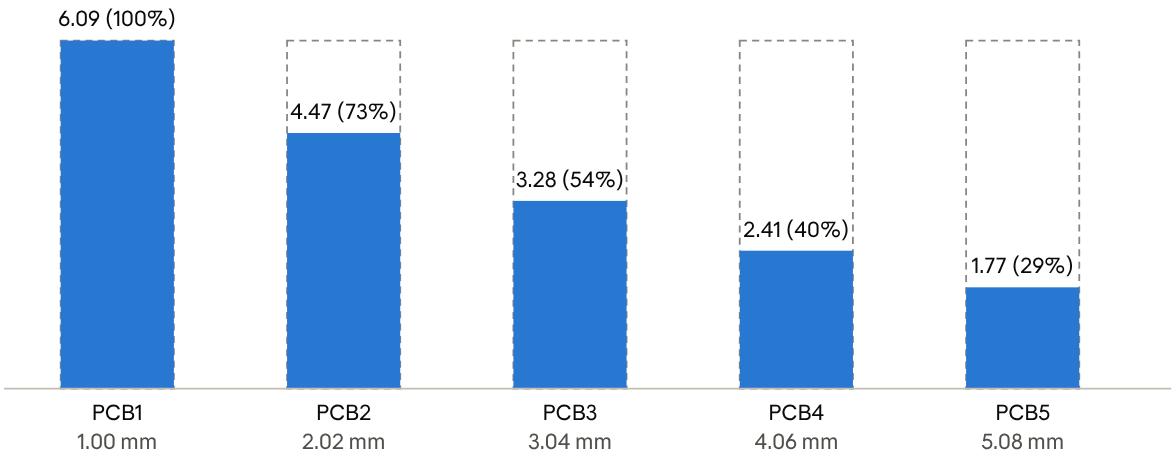
The stator has no iron, so plate thickness and air gap are independent and can be chosen separately.

Air gap and the five-board stack

The field from the magnets weakens by about 26% per mm of distance (k = 0.303 /mm). So at g = 1.0 mm the five-board stack gives **18.0 mN·m/A per phase, 2.96× one board** [AEDT + Py].

The five boards add up to about 3× one board, not 5×

Stack 18.0 mN·m/A per phase = 2.96× PCB1, 59% of the 30.4 if every board sat 1.0 mm from the magnets



Bars: each board's K1 in mN·m/A at its distance from the magnets (g = 1.0 mm). Dashed outline: the same board at 1.0 mm.

AEDT single-board K1 at 1.0 and 5.0 mm, exponential fit k = 0.3026 /mm; stack_sum.py, 2026-10-07

- **Single board at 5.0 mm** (REV0_DOE_D500) [AEDT]:
  - K1 = 1.815 mN·m/A, 29.8% of the value at 1 mm.
  - ψ = −30.0°, against −30.2° at 1 mm.
  - The waveform matches K·cos(θe + 30°) within about 3%, and the 3rd harmonic falls to 2.1%.
- **Decay rate.** k_eff = ln(6.09/1.815)/4 = 0.3026 /mm, 6% above the analytic π/τ of 0.286 /mm. That is expected: the shorter pole pitch near the inner radius makes the field fade faster.
- **Boards add in phase.** ψ barely moves with distance, so the phasor sum equals the plain sum to 99.99%.

Air gap g	Stack K (mN·m/A)	Ideal 3-phase torque (mN·m per A peak)	E1 at 1000 rpm (V rms)	Km (N·m/√W)
0.5 mm	21.0 (+16%)	31.5	1.55	0.0218
1.0 mm (selected)	**18.0**	**27.0**	**1.34**	**0.0187**
1.5 mm	15.5 (−14%)	23.2	1.15	0.0161

The 0.5 and 1.5 mm rows extrapolate the 1.0–5.0 mm fit. The 0.5 mm single-board run, once in, checks the fit. Km uses $R_\varphi = 1.389\ \Omega$, excluding the external leads.

Rev0 decision and final model

On 2026-10-07 Olwethu locked Rev0 at **g = 1.0 mm and t = 2.0 mm**. The final model REV0_FINAL_G100_T20_FROZEN.aedt is built and validated [AEDT].

- **Why 1.0 mm and not 0.5 mm.** A smaller gap always gives more torque, so the gap is set by what can be built:
  - PCB bow and twist (IPC-6012 allows up to 0.75%, about 0.7 mm across 93 mm before clamping; not yet checked against the standard);
  - rotor runout and bearing play;
  - magnet thickness tolerance and assembly error.

  1.0 mm is the validated baseline and keeps that margin. The 0.5 mm result is reported as the cost of this choice: about 16% torque.
- **How it was built.**
  - run_REV0_FINAL.py copied the frozen stator (it never opens the original) and re-probed all 36 junction points (0 failed).
  - It added the rotor and rebuilt the region with its +Y face on the lead ends.
  - AEDT validation passed with only the expected "no solution setup" notice. The run took 13.5 min.
- **Checked geometry.**
  - Stator −4.591 to +0.510 mm; magnet face at +1.510 mm (gap exactly 1.0000 mm); back plate top at +5.510 mm.
  - Rotor parts reach r = 43.0 mm, and the terminal leads start at r = 44.5 mm.

- **Ready for later tests.** The rotor angle is a variable (rot_angle) and the phase currents are variables (I_A, I_B, I_C = 0 A). The star point is left open. Tests run on copies of this file.
- **Where it is stored.** Motor run\frozen\ (OneDrive) and on PC 2′s local disk.

The full five-board motor is too large to solve on these PCs, so this model is the engineering geometry. The performance figures come from the single-board solves plus the stack sum.

Rev1 (S1–R1–S2–R2)

Rev1 doubles Rev0: two identical five-board stators in series, each driven by its own identical rotor. It is fully specified and scripted. Its numbers need one more single-board solve, REV1_S2_ENV, which takes about 50 min.

Item	Specification
Order	S1 → g → R1 magnets + steel → g → S2 → g → R2 magnets + steel
Rotors	Identical to Rev0 (16 × N35, 28 mm radius, 2 mm steel_1010); each rotor′s magnets face its own stator
Alignment	R2 at the same angle as R1 (0° offset); one common 19 mm shaft through the S2 bore
Stators	S2 is an exact copy of the frozen S1, shifted 11.102 mm along +Z (2g + 2 mm magnet + t + 5 pitches)
Connection	S1 and S2 in series per phase (10 boards), giving $R_\varphi \approx 2.777\ \Omega$; star point open
Open parameters	None: g = 1.0 mm and t = 2.0 mm, as for Rev0

- **What changes for S2.** S2′s last board sits 1 mm from R1′s steel plate, and that steel acts like a mirror for R2′s field. S2 therefore sees a stronger field than S1, mostly on the boards nearest R1.
- **How it is modelled.** Each S2 board′s result is multiplied by a first-order mirror-image factor, $1 + c \cdot e^{(-2k(L - d - p/2))}$, where $L = 2g + 5p$.

  - c is fitted to the REV1_S2_ENV solve: one board at g + 4 pitches from R2, with R1′s plate g below it.
  - Rev1 = K_S1 + K_S2.
- **What to report.** Rev1/Rev0 torque per amp and Km together. Resistance doubles, so near-doubled torque does not mean doubled efficiency. Before the extra solve, the estimate is a little above 2× the torque but only about 1.4–1.5× the Km [Est].
- **Scripts (ready).**
  - run_REV1_S2_ENV.py: the one FEM solve.
  - run_REV1_FINAL.py: builds and freezes the S1–R1–S2–R2 model, probes the S1 and S2 junctions, and adds S2′s terminals to the phase windings. Tested offline, not yet run in AEDT.
  - stack_sum.py: writes rev1_vs_rev0.csv.
- **Not modelled:** leakage from R1′s magnets through R1′s steel plate.

Assumptions, limits and risks

The biggest open risk is that the three-phase check has not yet passed. The biggest modelling limit is that the full motor is only computed, never solved in one piece.

Item	Type	Effect / mitigation
Three-phase synchronous check not passed (REF_ROTOR_MS_3PH)	Open check	All torque figures are provisional until 1.5·K1·I ≈ 9.1 mN·m at 1 A peak is confirmed
Stack sum fitted through two points (1.0 and 5.0 mm)	Method	The 0.5 mm point checks the exponential; if the fit misses it by more than about 2%, switch to interpolation
Stator copper and FR-4 do not change the rotor′s field (superposition)	Assump	Holds because the stator has no iron; armature reaction on the steel is neglected at ⩽ 1 A
FR-4 and mask as air	Assump	Verified: ⩽ 0.22% effect
Stack-up scaled from Altium ratios, not the JLCPCB stack-up	Assump	Moves layer heights by fractions of a mm; update once the real stack-up is supplied

Item	Type	Effect / mitigation
Vias as solid 0.30 mm copper	Assump	Fine for field and conduction; resistance is +2.9% against the reference
Torque from one current direction (+1 A) in the sweep runs	Method	Copper torque is linear in current; rotor virtual torque only a cross-check (it disagrees by up to 122% at 5 mm, which is expected)
Linearity (±0.5 A against ±1 A) and 5th-harmonic separation	Not yet checked	8 samples per period cannot separate the 5th harmonic from the 3rd
Inductance of the 5-board string	Estimate only	The 2-board run suggests a board-to-board mutual of about 1 µH; a full matrix freezes the PCs
Shaft material steel_1010	Assump	Affects only the field near the bore
Rev1: leakage from R1′s magnets neglected; first-order mirror model	Assump	Calibrated on one FEM point
Lab PCs: 16 GB, profiles may be wiped at logout	Practical	Every run checkpointed and restartable; results and frozen models copied to OneDrive

Next steps

Three solves and two short builds remain. Then the Rev0 and Rev1 numbers become final.

- [ ] Confirm whether REF_ROTOR_MS_3PH and REV0_DOE_D050 (0.5 mm) ran on PC 1 — send their rev0_run_log.txt files.
- [ ] Rerun REF_ROTOR_MS_3PH if it stopped early. It needs REV0_REF_ROTOR_MS_SWEEP_results in Motor run\results.
- [ ] PC 2: run_REV1_S2_ENV.py (about 50 min), then run_REV1_FINAL.py (about 30 min).
- [ ] Run stack_sum.py on all the results. It gives the final Rev0 numbers with the 0.5 mm check, plus rev1_vs_rev0.csv.
- [ ] Choose the star-point ends once the three-phase check confirms the polarity.

- [ ] Later: the ±0.5 A linearity check, and a magnet-radius study (27/28/29 mm) if time allows.

Sources:

- Vatani et al., *Analytical Modeling of Coreless AFPM Machines*, UK SPARK Lab, 2026: paper
- *PCB Motor Design Guide*, BestPCBs, 2026: guide
- Project records: rev0_motor_spec.md, results_2026-10-07.md, rev1_spec_and_final_builds.md; AEDT run logs, 2026-10-04 to 2026-10-07